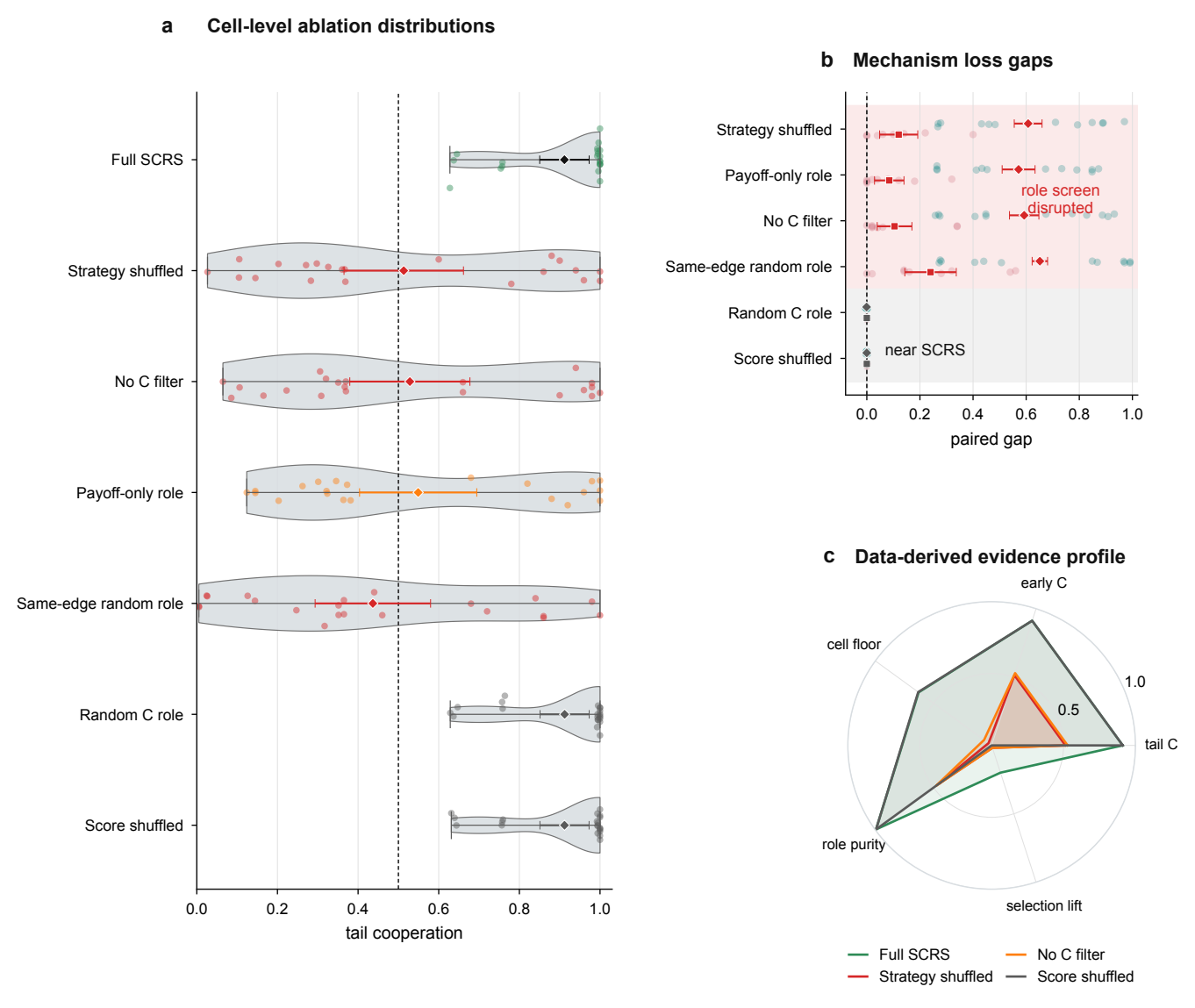

a
Cell-level ablation distributions
Full SCRS
Strategy shuffled
No C filter
Payoff-only role
Same-edge random role
Random C role
Score shuffled
0.0
0.2
0.4
0.6
0.8
1.0
tail cooperation
b
Mechanism loss gaps
Strategy shuffled
Payoff-only role
No C filter
Same-edge random role
Random C role
Score shuffled
role screen disrupted
near SCRS
0.0
0.2
0.4
0.6
0.8
1.0
paired gap
c
Data-derived evidence profile
early C
cell floor
1.0
0.5
tail C
role purity
selection lift
Full SCRS
No C filter
Strategy shuffled
Score shuffled